



Report on

RELIABILITY PREDICTION

OF

Camera Model No.: HC-IPC-DI3215DVZK-N5-O
(CCTV Camera)

as per Telcordia SR-332, Issue 4



CENTRE FOR RELIABILITY,
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TERMS AND CONDITIONS

1. This report refers only to the sample actually subjected to Design Reliability Analysis.
2. The analysis report provided is solely based on a theoretical estimation for predicting design reliability.
3. The accuracy and validity of the design reliability prediction cannot be guaranteed due to its theoretical nature.
4. The analysis report should be used for informational purposes only and should not be solely relied upon for making critical decisions.
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1.0 SCOPE

- 1.1** Service request no. : 2602328
- 1.2** Name & Address of the Client : Hi-Focus Electronics India Pvt Ltd
3, Sunguvar Agraharam St,
Chindatripet, Chintadripet, Chennai,
Tamil Nadu 600002.
- 1.3** Description : **CCTV CAMERA**
Make : Hi-Focus Electronics India Pvt Ltd
Model No. : HC-IPC-DI3215DVZK-N5-O
Serial No. : HFP2510054312000286
- 1.4** Project : Received Date : 26-03-2026
Completed Date : 21-04-2026
- 1.5** (a) Standard/ Analysis / Test Procedure Ref. : Telcordia SR-332, Issue 4
(b) Scope : Reliability Prediction by Parts Count Method

Approved by

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2.0 INTRODUCTION

This report presents the results of Reliability Prediction Analysis performed on **Camera Model No.: HC-IPC-DI3215DVZK-N5-O** as per Telcordia SR-332, Issue 4 as depicted in Figure 1.0, 2.0, 3.0, 4.0, 5.0, and 6.0. This exercise was performed under **Ground, Fixed, Controlled (GB)** environment as requested by **Hi-Focus Electronics India Pvt Ltd.**

Design Reliability Prediction is the process of quantitatively assessing whether a proposed or actual equipment design will meet a specified reliability requirement. Design Reliability Prediction has various applications in the design and development process including substantiating quantitative design comparisons with competitive products, selecting warranty periods, advertising and marketing. It provides guidance, relative to expected inherent reliability of a given design.



Figure 1.0 (a & b) Front & Top view of camera



Figure 2.0 (a) Markings on camera

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**Reliability Prediction Report on
Camera Model No.: HC-IPC-DI3215DVZK-N5-O**

**Date of Issue
21-04-2026**

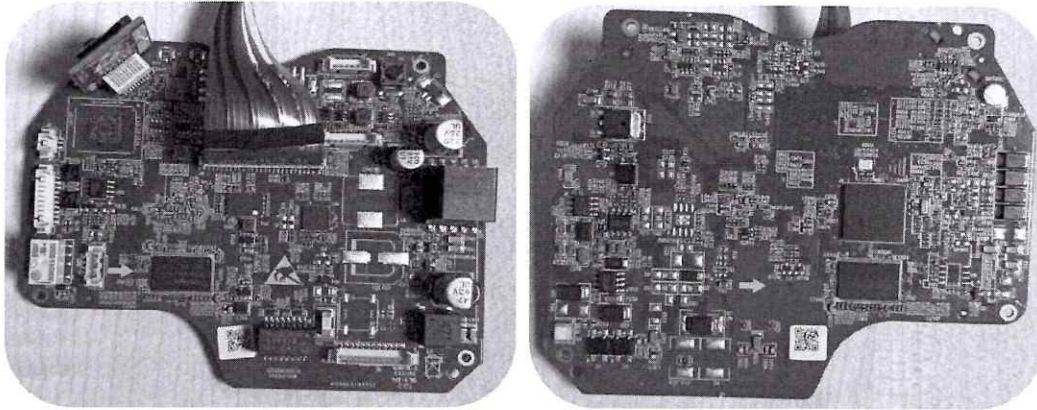


Figure 3.0 (a & b) Front & Rear View of PCB 1

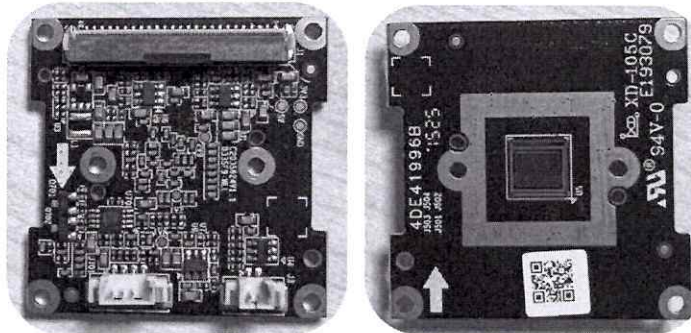


Figure 4.0 (a & b) Front & Rear View of PCB 2



Figure 5.0 (a) Motorized Lens



Figure 6.0 (a & b) IR Board & LEDs

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3.0 APPLICABLE DOCUMENTS AND TOOLS

Telcordia SR-332, Issue 4 - Reliability Prediction Procedure for Electronic Equipment

4.0 RELIABILITY PREDICTION OBJECTIVES

Objectives of this analysis are to estimate the Mean Time between Failure (MTBF) of **Camera Model No.: HC-IPC-DI3215DVZK-N5-O** using Telcordia SR-332, Issue 4 and to obtain quantitative design information as applicable. The main objectives are:

- a. Provide an estimate of the steady-state failure rates for device design based on generic reliability parameters and obtain feasibility evaluation.
- b. Provide feedback for designers.
- c. Identify potential problem areas during design review.
- d. Provide MTBF of the system and sub-assemblies and failure rate of system/sub-assemblies/ Components.

5.0 SR-332, Method 1: Unit Steady-State Failure Rate Parts Count Method - ASSUMPTIONS

Reliability Prediction using SR-332, Method 1: Unit Steady-State Failure Rate Parts Count Method for reliability prediction operates under the following assumptions: assumes a serial relationship between parts of the system, in which, failure of any part constitutes system failure. The model also assumes that

- a. At the device level, it is assumed that no detailed reliability data is available. Predictions are based on generic reliability parameters.
- b. Predictions are made using generic reliability parameters as specified in the relevant sections of SR-332, which are applied to estimate steady state failure rates.
- c. At the unit level, failure rate predictions are based solely on the failure rates of the individual devices that compose the unit, without considering interactions between devices.
- d. This method primarily estimates steady-state failure rates and does not specifically account for infant mortality failures (early-life failures) or wear out failures (end-of-life failures). It assumes a constant failure rate.

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6.0 SR-332, Method I: Unit Steady-State Failure Rate Parts Count Method: Factors

Method I: Unit Steady-State Failure Rate Using the Parts Count Method primarily estimates reliability by calculating the steady-state failure rate of a device using a set of generic parameters. It takes into account the following factors:

- 6.0.1 Number of component of each Type
- 6.0.2 Mean Generic Steady-State Failure Rate
- 6.0.3 Temperature factor
- 6.0.4 Environmental factor
- 6.0.5 Quality factor
- 6.0.6 Electrical Stress Factor

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7.0 ENVIRONMENTAL CONDITIONS

7.1 STORAGE AND OPERATIONAL ENVIRONMENT

The Reliability Prediction has been performed using **Ground, Fixed, Controlled (GB)** Environment based on the operational conditions of **Camera Model No.: HC-IPC-DI3215DVZK-N5-O**.

GROUND, FIXED, CONTROLLED (GB):

The environment represents a stationary, land-based installation located in a protected or climate-controlled facility. It is characterized by low vibration and shock stresses, low atmospheric variations, and low temperature-cycling stresses, with the understanding that a building or ruggedized enclosure provides consistent protection from weather and contaminants.

TEMPERATURE

Reliability Prediction has been performed on **Camera Model No.: HC-IPC-DI3215DVZK-N5-O** at the **Ambient Temperature of 40 °C** for **Ground, Fixed, Controlled (GB)** Environment.

7.2 QUALITY LEVEL : Level- 1

The unit has been evaluated under the commercial-grade component Quality level-1 as mentioned by the client in bill of material. The components used in the unit were procured without undergoing thorough device qualification or lot-to-lot controls by the equipment manufacturer.

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8.0 REPORTS

Report includes the following information-

- a) Components/ Parts descriptions
- b) Failure Rate
- c) Pi (π) Factors
- d) Parameters details

Refer Annexure- I.

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9.0 System Analysis Summary

The overall estimated MTBF and Failure Rate as per Telcordia SR-332 of **Camera Model No.: HC-IPC-DI3215DVZK-N5-O**, submitted for reliability prediction analysis by **Hi-Focus Electronics India Pvt Ltd** are summarized as follows:

SYSTEM: Camera Model No.: **HC-IPC-DI3215DVZK-N5-O**

Environment	Ambient Temperature	Failure Rate (Failures/Million Hours)	MTBF (Hrs)
Ground, Fixed, Controlled (GB)	40°C	4.135917	241784.35

- Design reliability prediction analysis was carried out based on the failure rates of components used in the camera (refer Annexure I), following a physical verification of components and in accordance with Telcordia SR-332, Issue 4.
- The camera contains three (03) nos of synchronous/stepper/shutter motors. However, failure rate data for these motors is not available in the Telcordia standard or in the reliability prediction software used. Therefore, for the purpose of the design reliability prediction, the failure rate for each motor has been conservatively assumed as one (1) FIT.
- MTBF (Mean Time Between Failures) calculations have been performed using the component details—such as type, quantity, electrical and environmental parameters, stress levels, and quality grades—as provided by the client in their submitted Bill of Materials.

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10.0 Abbreviation Used:

- a) MTBF : Mean Time Between Failure
- b) λG : Base failure rate for the generic part (Ground fixed base rate)
- c) σG : Standard deviation or uncertainty in base failure rate (generic)
- d) πQ : Quality factor (related to part/process quality level)
- e) πE : Environmental factor (based on the operating environment, e.g., ground benign, airborne)
- f) πT : Temperature factor (influenced by ambient/operating temperature)
- g) πS : Stress factor (e.g., voltage or power stress ratio)
- h) λSS : Subsystem or system-level failure rate (or stress-adjusted failure rate)
- i) σSS : Standard deviation or uncertainty in system/subsystem failure rate

- End of Report -

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Annexure - I

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Reliability Prediction Report									
Name			CCTV Camera			Calculation Model			
Description			Make : HI-FOCUS			Environment			
			Model No : HC-IPC-DI3215DVZK-N5-O			Ground, Fixed, Controlled (GB)			
			Serial No. : HFP2510054312000286			Ambient Temperature, Celsius			
			Job No : 10TE202600689			Operating Profile			
Quantity 1						Telecommunications			
						Failure Rate Units			
Reference Designator						Failures/Million Hours			
						Failure Rate			
						4.135917			
						Hours			
						MTBF Units			
						MTBF 241784.35			

PARTS

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.1	R0402RXX33 0XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R3,R7,R8,R10,R11,R12,R13	330 $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	7	0.000173	0.001211	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.2	R0402RXX47 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R90,R91	4.7K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 4.7, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.3	R0402RXX33 0XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R3,R4,R18,R30,R57,R58	330 $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	6	0.000175	0.001048	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)
 Reviewed By: Shubham Kumar (Scientist D)
 Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.4	R0402RXX20 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1,R14,R48,R71 4	2K $\pm$ 1%-0402-(-55°C~155°C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.5	R0402RXX22 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R52	2.2K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2.2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.6	R0402RXX10 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R17,R716	1K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.7	R0402RXX10 5XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R702	1M $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: Mohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.8	R0402RX12 1XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R20	1200 $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 120, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.9	R0402RX10 3XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R9,R10,R631,R632,R704,R705,R712,R715	10K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	8	0.000172	0.001374	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.10	R0402RX10 4XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R701,R703,R709	100K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	3	0.000183	5.501810E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.11	R0402RX24 1XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R713	240 $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 240, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.12	R0402JXX00 00P16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R11,R47,R49,R625,R708	00 $\pm$ 5%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	5	0.000147	7.328089E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.53301804, λ G: 0.08, σ G: 0.02, λ SS: 0.12792433, σ SS: 0.031981083, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.13	R0402RXX18 3XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R707, R710, R711	18K $\pm$ 1%-0402-(-55°C~155°C)	3	0.000183	5.501810E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 18, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.14	R0402RXX15 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R7, R717	1.5K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1.5, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.15	R0402RXX33 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R50	3.3K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 3.3, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.16	R0402RXX47 3XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R706	47K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 47, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.17	R0402RXX75 1XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R12	7500 $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 750, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.18	R0402RXX15 3XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R13	15K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 15, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.19	R0402RXX47 1XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R706, C1401	4700 $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 470, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.20	R0402RXX47 4XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R41, R1419	470K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 470, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)
Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.21	R0402RXX47 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R45, R52, R63, R86, R87, R90, R93, R124, R150, R191, R220, R227, R274, R277, R703, R712, R737, R742, R746, R803, R809, R812, R815, R818, R822, R823, R824, R859, R860, R863, R897, R1301, R1402	4.7K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	33	0.000163	0.005374	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 4.7, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.22	R0402RXX33 0XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R5, R6, R10, R11, R17, R18, R25, R61, R94, R101, R153, R157, R253, R517, R518, R804, R811, R825, R857, R858, R861, R864, R865, R866, R867, R868	33 Ω $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	26	0.000164	0.004263	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 33, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.23	R0402RXX20 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R20, R91, R99, R140, R181, R183, R186, R188, R246, R896, R1403	2K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	11	0.000169	0.001861	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.24	R0402RXX22 3XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R141, R724	22K $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 22, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.25	R0402RXX22 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R78,R603	2.2K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2.2, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.26	R0402RXX10 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R21,R23,R27,R1 33,R134,R138,R 147,R149,R151, R171,R217,R248 ,R345,R734,R74 4,R748,R830,R1 003,R1005	1K $\pm$ 1%-0402-(-55°C~155°C)	19	0.000166	0.003146	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 1, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.27	R0402RXX10 3XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R3,R29,R50,R10 0,R130,R143,R1 45,R219,R236,R 238,R239,R247, R601,R602,R709 ,R722,R736,R73 8,R740,R741,R8 31,R1013,R1015 ,R1016,R1201,R 2621	10K $\pm$ 1%-0402-(-55°C~155°C)	26	0.000164	0.004263	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.28	R0402RXX10 4XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R721,R801	100K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.29	R0402RXX24 1XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1007,R1008	2400 $\pm$ 1%-0402-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 240, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.30	CR-05JL7--- 10R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R113,R114,R115,R116,R120,R26,2,R899	100 $\pm$ 5%-0805-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	7	0.000156	0.001091	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.57926223, λ G: 0.08, σ G: 0.02, λ SS: 0.13902294, σ SS: 0.034755734, Temperature: 40
1.31	CR-05FL7--- 10K	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R117,R118	10K $\pm$ 1%-0805-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	2	0.000171	3.426031E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.57926223, λ G: 0.08, σ G: 0.02, λ SS: 0.13902294, σ SS: 0.034755734, Temperature: 40
1.32	CR-05FL7-- 100R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R178	1000 $\pm$ 1%-0805-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	1	0.000185	1.850080E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Applied Power: 0.01, Rated Power: 0.125, Percentage: 8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.57926223, λ G: 0.08, σ G: 0.02, λ SS: 0.13902294, σ SS: 0.034755734, Temperature: 40
1.33	CR-05JL7--- OR	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R8,R35,R107,R112	00 $\pm$ 5%-0805-(-55 $^{\circ}$ C~155 $^{\circ}$ C)	4	0.000147	5.890311E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Applied Power: 0.001, Rated Power: 0.125, Percentage: 0.8	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.52750338, λ G: 0.08, σ G: 0.02, λ SS: 0.12660081, σ SS: 0.031650203, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.34	R0402JXX00 00P16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R2,R13,R16,R76, R160,R166,R240 ,R637,R646,R64 7,R650,R651,R6 65,R723,R735,R 739,R750,R756, R766,R819,R826 ,R827,R833,R10 01,R1002,R1203 ,R1423,R1599,R 1705,R1723	100±5%-0402-(- 55°C~155°C)	30	0.000135	0.004062	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0, Resistance Units: ohm, Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Applied Power: 0.001, Rated Power: 0.0625, Percentage: 1.6	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.53301804, λ G: 0.08, σ G: 0.02, λ SS: 0.12792433, σ SS: 0.031981083, Temperature: 40
1.35	R0402RXX10 0XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R121,R229,R230 ,R231,R233,R23 4,R235,R2622	100±1%-0402-(- 55°C~155°C)	8	0.000172	0.001374	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 10, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.36	R0402RXX10 1XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R14,R19,R24,R2 8,R814,R817,R1 404	1000±1%-0402-(- 55°C~155°C)	7	0.000173	0.001211	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 100, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.37	CR-03FL7--- 75R	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R290,R291,R421 ,R422	750±1%-0603-(- 55°C~155°C)	4	0.000535	0.002139	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 75, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.01, Percentage: 100	Applied Power: 0.01, Rated Power: 0.01, Percentage: 100	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 1.9155408, λ G: 0.08, σ G: 0.02, λ SS: 0.4597298, σ SS: 0.11493245, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

Annexure I

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.38	R0402RXX18 3XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R84, R729, R1110	18K $\pm$ 1%-0402-(-55°C~155°C)	3	0.000183	5.501810E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.39	R0402RXX33 3XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R743, R745	33K $\pm$ 1%-0402-(-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.40	R0402RXX15 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R237	1.5K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.41	R0402RXX33 2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R142, R170, R224, R717, R832, R898	3.3K $\pm$ 1%-0402-(-55°C~155°C)	6	0.000175	0.001048	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40



Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)

Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.42	R0402RXX22 4XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R128	220K $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 220, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.43	R0402RXX20 1XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R139	2000 $\pm$ 1%-0402-(-55°C~155°C)	1	0.000205	2.052850E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 200, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.44	R0402RXX15 3XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R155, R212, R218, R704, R720, R725, R732, R747	15K $\pm$ 1%-0402-(-55°C~155°C)	8	0.000172	0.001374	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 15, Resistance Units: Kohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.45	R0402RWX2 R2XF16LTA	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R273, R286, R287, R289	2.20 $\pm$ 1%-0402-(-55°C~155°C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 2.2, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.46	RL1206FR- 070R51L	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R123, R127, R131, R132	510m Ω $\pm$ 1%-1206-(-55°C~155°C)	4	0.000154	6.140516E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance: 0.51, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.25, Percentage: 4	Applied Power: 0.01, Rated Power: 0.25, Percentage: 4	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.54991036, λ G: 0.08, σ G: 0.02, λ SS: 0.13197849, σ SS: 0.032994621, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.47	RS-05FL7---OR2	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R81	200mR $\pm$ 1%-0805 (-55°C ~155°C)	1	0.000280	2.804510E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: ohm, Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Applied Power: 0.05, Rated Power: 0.125, Percentage: 40	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.87809543, λ G: 0.08, σ G: 0.02, λ SS: 0.2107429, σ SS: 0.052685726, Temperature: 40
1.48	CR0402FF56 ROG	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1014, R1017	56R $\pm$ 1%-0402 (-55°C~155°C)	2	0.000190	3.801525E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.49	CR0402FF68 00G	Resistor	Fixed	Film (Carbon, Oxide, Metal)	R1018, R1019, R1020, R1021	680R $\pm$ 1%-0402 (-55°C~155°C)	4	0.000179	7.177196E-004	Type: Film (Carbon, Oxide, Metal), Quality Level: Level I, Resistance Units: ohm, Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Applied Power: 0.01, Rated Power: 0.0625, Percentage: 16	Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, π S: 0.64274964, λ G: 0.08, σ G: 0.02, λ SS: 0.15425991, σ SS: 0.038564978, Temperature: 40
1.50	CC0805KKX5 R7BB106	Capacitor	Ceramic	General	C1	10uF $\pm$ 10%-16V-X5R-0805	1	0.000102	1.016686E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.29987937, λ G: 0.1, σ G: 0.01, λ SS: 0.08996381, σ SS: 0.008996381, Temperature: 40
1.51	CC0402KRX7 R7BB104	Capacitor	Ceramic	General	C2	100nF $\pm$ 10%-16V-X7R-0402	1	0.000102	1.016686E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.29987937, λ G: 0.1, σ G: 0.01, λ SS: 0.08996381, σ SS: 0.008996381, Temperature: 40
1.52	CC0402IRNP O9BN100	Capacitor	Ceramic	General	C1	10pF $\pm$ 5%-50V-NPO-0402	1	0.000055	5.526646E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 50, Percentage: 5.8	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16329449, λ G: 0.1, σ G: 0.01, λ SS: 0.048988348, σ SS: 0.0048988348, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.53	CC0402KRX7 R98B102	Capacitor	Ceramic	General	C6,C31,C47,C50, C54,C658	1nF $\pm$ 10%-50V- X7R-0402	6	0.000045	2.690611E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.14204662, λ G: 0.1, σ G: 0.01, λ SS: 0.042613985, σ SS: 0.0042613985, Temperature: 40
1.54	0402X105K6 R3CT	Capacitor	Ceramic	General	C14,C40,C81,C7 12	1uF $\pm$ 10%-6.3V- X5R-0402	4	0.000090	3.589336E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 6.3, Percentage: 19.0476190476	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.28109921, λ G: 0.1, σ G: 0.01, λ SS: 0.084329762, σ SS: 0.0084329762, Temperature: 40
1.55	CC0402KRX7 R78B104	Capacitor	Ceramic	General	C2,C9,C11,C16, C37,C63,C64,C7 0,C74,C77,C82,C 83,C84,C86,C98, C457,C657,C660 ,C661,C668,C68 6,C701,C702,C7 03,C704,C706,C 707,C708,C709, C711	100nF $\pm$ 10%- 16V-X7R-0402	30	0.000092	0.002762	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 16, Percentage: 20.625	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.29987937, λ G: 0.1, σ G: 0.01, λ SS: 0.08996381, σ SS: 0.008996381, Temperature: 40
1.56	0402B103K5 00CT	Capacitor	Ceramic	General	C5,C48	10nF $\pm$ 10%-50V- X7R-0402	2	0.000046	9.295127E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	AC Peak Voltage: 0, Applied DC Voltage: 1.2, Rated Voltage: 50, Percentage: 2.4	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.14204662, λ G: 0.1, σ G: 0.01, λ SS: 0.042613985, σ SS: 0.0042613985, Temperature: 40
1.57	0402X225M 6R3CT	Capacitor	Ceramic	General	C85,C687	2.2uF $\pm$ 20%- 6.3V-X5R-0402	2	0.000278	5.561161E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.84984745, λ G: 0.1, σ G: 0.01, λ SS: 0.25495424, σ SS: 0.025495424, Temperature: 40



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 Reviewed By: Shubham Kumar (Scientist D)
 Approved By: V. Jambulingam (Head Reliability)

Annexure I

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.58	CL05A475M Q5NRNC	Capacitor	Ceramic	General	C75,C76,C78,C79,C80,C87,C663,C664,C666,C669,C670	4.7uF±20%-6.3V-X5R-0402	11	0.000265	0.002913	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	AC Peak Voltage: 0, Applied DC Voltage: 2.9, Rated Voltage: 6.3, Percentage: 46.0317460317	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.84984745, λ G: 0.1, σ G: 0.01, λ SS: 0.25495424, σ SS: 0.025495424, Temperature: 40
1.59	CL10A106KP 8NNNC	Capacitor	Ceramic	General	CL15,C38,C436,C705,C710	10uF±10%-10V-X5R-0603	5	0.000317	0.001586	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	π Q: 3, π E: 1, π T: 1, π S: 1, λ G: 0.1, σ G: 0.01, λ SS: 0.3, σ SS: 0.03, Temperature: 40
1.60	CL10A226M Q8NRNC	Capacitor	Ceramic	General	C3,C4,C8,C10,C12,C17,C18,C20,C29,C30,C33,C35,C36,C66,C67,C71,C72,C88,C105	22uF±20%-6.3V-X5R-0603	19	0.000340	0.006469	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	π Q: 3, π E: 1, π T: 1, π S: 1.1025427, λ G: 0.1, σ G: 0.01, λ SS: 0.33076281, σ SS: 0.033076281, Temperature: 40
1.61	CC0402JRNPO9BN100	Capacitor	Ceramic	General	C825	10pF±5%-50V-NP0-0402	1	0.000057	5.710925E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.62	CC0402JRNPO9BN180	Capacitor	Ceramic	General	C815,C822,C824	18pF±5%-50V-NP0-0402	3	0.000054	1.631020E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.63	CC0402JRNPO9BN101	Capacitor	Ceramic	General	C181,C991	100pF±5%-50V-NP0-0402	2	0.000063	1.2693350E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.19398004, λ G: 0.1, σ G: 0.01, λ SS: 0.058194013, σ SS: 0.0058194013, Temperature: 40



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Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.64	CC0402KRX7R9BB102	Capacitor	Ceramic	General	C182,C1408,C1417	1nF $\pm$ 10%-50V-X7R-0402	3	0.000054	1.631020E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, Δ G: 0.1, σ G: 0.01, Δ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.65	0402B332K500CT	Capacitor	Ceramic	General	C73	3.3nF $\pm$ 10%-50V-X7R-0402	1	0.000057	5.710925E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, Δ G: 0.1, σ G: 0.01, Δ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.66	CC1206KKX7RDBB102	Capacitor	Ceramic	General	C30,C89,C126,C158,C167,C168,C284,C327	1nF $\pm$ 10%-2KV-X7R-1206	8	0.000041	3.262902E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 2000, Percentage: 0.25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.13006122, Δ G: 0.1, σ G: 0.01, Δ SS: 0.039018367, σ SS: 0.0039018367, Temperature: 40
1.67	0402N221J500CT	Capacitor	Ceramic	General	C171,C173,C705,C716,C731	220pF $\pm$ 5%-50V-NPO-0402	5	0.000054	2.676154E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, Δ G: 0.1, σ G: 0.01, Δ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40
1.68	1206B104K201NT	Capacitor	Ceramic	General	C159,C166	100nF $\pm$ 10%-200V/250V-X7R-1206	2	0.000113	2.253547E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 200, Percentage: 24	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 200, Percentage: 24	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.3443833, Δ G: 0.1, σ G: 0.01, Δ SS: 0.10331499, σ SS: 0.010331499, Temperature: 40
1.69	1206B105K101CT	Capacitor	Ceramic	General	C161,C163	1uF $\pm$ 10%-100V-X7R-1206	2	0.000301	6.028542E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 100, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 100, Percentage: 48	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.92127196, Δ G: 0.1, σ G: 0.01, Δ SS: 0.27638159, σ SS: 0.027638159, Temperature: 40
1.70	CC0805KKX5R7BB106	Capacitor	Ceramic	General	C58,C62	10uF $\pm$ 10%-16V-X5R-0805	2	0.000912	0.001824	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 16, Percentage: 75	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 16, Percentage: 75	Level I	π Q: 3, π E: 1, π T: 1, π S: 2.7870955, Δ G: 0.1, σ G: 0.01, Δ SS: 0.83612864, σ SS: 0.083612864, Temperature: 40



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Approved By: V. Jambulingam (Head Reliability)

SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.71	VZH221M1E TR-0810	Capacitor	Electrolytic	Aluminum	CE4	220uF±20% 25V-105°C- Ø8*10mm	1	0.002730	0.002730	Type: Aluminum, Quality Level: Level I, Capacitance: 220, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.95313379, ΛG : 0.73, σG : 0.17, ΛSS : 2.087363, σSS : 0.48609823, Temperature: 40
1.72	0603N821J5 00CT	Capacitor	Ceramic	General	C282, C283, C325 , C326	820pF±5%-50V- NPO-0603	4	0.000054	2.154621E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	πQ : 3, πE : 1, πT : 1, πS : 0.16873936, ΛG : 0.1, σG : 0.01, ΛSS : 0.050621808, σSS : 0.0050621808, Temperature: 40
1.73	0402X105K6 R3CT	Capacitor	Ceramic	General	C208, C739, C801 , C821	1uF±10%-6.3V- X5R-0402	4	0.000352	0.001408	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	πQ : 3, πE : 1, πT : 1, πS : 1.1025427, ΛG : 0.1, σG : 0.01, ΛSS : 0.33076281, σSS : 0.033076281, Temperature: 40



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Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.74	CC0402KRX7R7B104	Capacitor	Ceramic	General	C28,C47,C50,C53,C57,C68,C88,C91,C145,C172,C175,C176,C180,C207,C711,C715,C719,C726,C733,C738,C741,C747,C751,C753,C802,C804,C808,C810,C813,C814,C817,C818,C819,C820,C823,C827,C829,C830,C831,C832,C835,C836,C838,C850,C852,C853,C854,C855,C858,C859,C860,C983,C984,C992,C995,C1002,C1003,C1004,C1006,C1012,C1013,C1014,C1015,C1016,C1017,C1018,C1019,C1020,C1021,C1022,C1023,C1024,C1025,C1026,C1028,C1029,C1030,C1031,C1032,C1033,C1034,C1035,C1036,C1037,C1038,C1039,C1040,C1041,C1102	10nF $\pm$ 10%-16V-X7R-0402	89	0.000141	0.012546	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 16, Percentage: 31.25	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 16, Percentage: 31.25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.4635922, λ G: 0.1, σ G: 0.01, λ SS: 0.13907766, σ SS: 0.013907766, Temperature: 40
1.75	0402B103K500CT	Capacitor	Ceramic	General	C183,C280,C281,C601,C1042	10nF $\pm$ 10%-50V-X7R-0402	5	0.000054	2.676154E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 50, Percentage: 6.6	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 50, Percentage: 6.6	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.16873936, λ G: 0.1, σ G: 0.01, λ SS: 0.050621808, σ SS: 0.0050621808, Temperature: 40

Tested and Prepared By: Shantanu Paria (TMB) and Anupam Anand (Scientist B)
Reviewed By: Shubham Kumar (Scientist D)
Approved By: V. Jambulingam (Head Reliability)



Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.76	0402X225M 6R3CT	Capacitor	Ceramic	General	C186,C1461	2.2uF±20%- 6.3V-X5R-0402	2	0.000361	7.214727E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	πQ: 3, πE: 1, πT: 1, πS: 1.1025427, λG: 0.1, σG: 0.01, λSS: 0.33076281, σSS: 0.033076281, Temperature: 40
1.77	0805X226M 100CT	Capacitor	Ceramic	General	C36	22uF±20%-10V- 0805	1	0.000338	3.384465E-004	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	πQ: 3, πE: 1, πT: 1, πS: λG: 0.1, σG: 0.01, λSS: 0.3, σSS: 0.03, Temperature: 40
1.78	UL1J470M0 810VG	Capacitor	Electrolytic	Aluminum	CE5	47uF±20%-63V- 105°C- Ø8*10mm	1	0.005369	0.005369	Type: Aluminum, Quality Level: Level I, Capacitance: 47, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 63, Percentage: 76.1904761905	AC Peak Voltage: 0, Applied DC Voltage: 48, Rated Voltage: 63, Percentage: 76.1904761905	Level I	πQ: 3, πE: 1, πT: 1, πS: 1.8749302, λG: 0.73, σG: 0.17, λSS: 4.1060971, σSS: 0.9562144, Temperature: 40
1.79	UT1E101M0 607VG	Capacitor	Electrolytic	Aluminum	CE3	100uF±20%- 25V-105°C- Ø6.3*7.7mm	1	0.002730	0.002730	Type: Aluminum, Quality Level: Level I, Capacitance: 100, Capacitance Units: uF, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	πQ: 3, πE: 1, πT: 1, πS: 0.95313379, λG: 0.73, σG: 0.17, λSS: 2.087363, σSS: 0.48609823, Temperature: 40
1.80	CL05A475M Q5NRNC	Capacitor	Ceramic	General	C209,C805,C807 ,C809,C811,C81 6,C828,C857,C1 402,C1403	4.7uF±20%- 6.3V-X5R-0402	10	0.000344	0.003442	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	πQ: 3, πE: 1, πT: 1, πS: 1.1025427, λG: 0.1, σG: 0.01, λSS: 0.33076281, σSS: 0.033076281, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.81	CL10A106KP8NNC	Capacitor	Ceramic	General	C109,C704,C714,C728,C735,C746,C812,C826,C851,C856,C985,C989,C993,C1001,C1027	10 μ F $\pm$ 10%-10V-X5R-0603	15	0.000310	0.004649	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	AC Peak Voltage: 0, Applied DC Voltage: 5, Rated Voltage: 10, Percentage: 50	Level I	π Q: 3, π E: 1, π T: 1, π S: 1, λ G: 0.1, σ G: 0.01, λ SS: 0.3, σ SS: 0.03, Temperature: 40
1.82	CL21A106KAYNNNE	Capacitor	Ceramic	General	C63,C92,C156,C177,C178,C179,C743	10 μ F $\pm$ 10%-25V-X5R-0805	7	0.000290	0.002028	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	AC Peak Voltage: 0, Applied DC Voltage: 12, Rated Voltage: 25, Percentage: 48	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.92127196, λ G: 0.1, σ G: 0.01, λ SS: 0.27638159, σ SS: 0.027638159, Temperature: 40
1.83	CL10A226M08NRNC	Capacitor	Ceramic	General	C29,C70,C706,C710,C717,C730,C732,C737,C750,C803	22 μ F $\pm$ 20%-6.3V-X5R-0603	10	0.000344	0.003442	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	AC Peak Voltage: 0, Applied DC Voltage: 3.3, Rated Voltage: 6.3, Percentage: 52.380952381	Level I	π Q: 3, π E: 1, π T: 1, π S: 1.1025427, λ G: 0.1, σ G: 0.01, λ SS: 0.33076281, σ SS: 0.033076281, Temperature: 40
1.84	HCM1206X7R103K102PT	Capacitor	Ceramic	General	C169	10nF $\pm$ 10%-1000V-X7R-1206	1	0.000044	4.356988E-005	Type: General, Quality Level: Level I, AC Peak Voltage: 0, Applied DC Voltage: 0, Rated Voltage: 1000, Percentage: 0	AC Peak Voltage: 0, Applied DC Voltage: 0, Rated Voltage: 1000, Percentage: 0	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.1287349, λ G: 0.1, σ G: 0.01, λ SS: 0.038620471, σ SS: 0.0038620471, Temperature: 40
1.85	C1402-1009-211	Connection	Multi-Pin		J2	193-F093K230-B06	1	0.001597	0.001597	Quality Level: Level I, Number of Pins: 10		Level I	λ G: 0.4, σ G: 0.1, π Q: 3, π E: 1, π T: 1, λ SS: 1.2, σ SS: 0.3, Temperature: 40
1.86	510-628-11	Connection	Multi-Pin		J6	2*10*0.8mm-5.2mm	1	0.003194	0.003194	Quality Level: Level I, Number of Pins: 20		Level I	λ G: 0.8, σ G: 0.2, π Q: 3, π E: 1, π T: 1, λ SS: 2.4, σ SS: 0.6, Temperature: 40
1.87	1.25T-1-LT-4A	Connection	Multi-Pin		J701	4*1.25mm	1	0.000639	6.387712E-004	Quality Level: Level I, Number of Pins: 4		Level I	λ G: 0.16, σ G: 0.04, π Q: 3, π E: 1, π T: 1, λ SS: 0.48, σ SS: 0.12, Temperature: 40
1.88	1.25T-1-LT-2A	Connection	Multi-Pin		J3	2*1.25mm	1	0.000319	3.193856E-004	Quality Level: Level I, Number of Pins: 2		Level I	λ G: 0.08, σ G: 0.02, π Q: 3, π E: 1, π T: 1, λ SS: 0.24, σ SS: 0.06, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.89	60076-5007-1001	Connection	Multi-Pin		J1	50*0.5mm-Flip-Latch	1	0.007985	0.007985	Quality Level: Level I, Number of Pins: 50		Level I	λG : 2, σG : 0.5, πQ : 3, πE : 1, πT : 1, λSS : 6, σSS : 1.5, Temperature: 40
1.90	1.25T-1-LT-10A	Connection	Multi-Pin		J6	10*1.25mm	1	0.001597	0.001597	Quality Level: Level I, Number of Pins: 10		Level I	λG : 0.4, σG : 0.1, πQ : 3, πE : 1, πT : 1, λSS : 1.2, σSS : 0.3, Temperature: 40
1.91	1.25T-1-LT-8A	Connection	Multi-Pin		J11	8*1.25mm	1	0.001278	0.001278	Quality Level: Level I, Number of Pins: 8		Level I	λG : 0.32, σG : 0.08, πQ : 3, πE : 1, πT : 1, λSS : 0.96, σSS : 0.24, Temperature: 40
1.92	1.25T-1-LT-6A	Connection	Multi-Pin		J3	6*1.25mm	1	0.000958	9.581568E-004	Quality Level: Level I, Number of Pins: 6		Level I	λG : 0.24, σG : 0.06, πQ : 3, πE : 1, πT : 1, λSS : 0.72, σSS : 0.18, Temperature: 40
1.93	1.25T-1-LT-2A	Connection	Multi-Pin		J8	2*1.25mm	1	0.000319	3.193856E-004	Quality Level: Level I, Number of Pins: 2		Level I	λG : 0.08, σG : 0.02, πQ : 3, πE : 1, πT : 1, λSS : 0.24, σSS : 0.06, Temperature: 40
1.94	1.25T-1-LT-12A	Connection	Multi-Pin		J5	12*1.25mm	1	0.001916	0.001916	Quality Level: Level I, Number of Pins: 12		Level I	λG : 0.48, σG : 0.12, πQ : 3, πE : 1, πT : 1, λSS : 1.44, σSS : 0.36, Temperature: 40
1.95	510-627-11	Connection	Multi-Pin		J2	2*10*0.8mm-5.2mm	1	0.003194	0.003194	Quality Level: Level I, Number of Pins: 20		Level I	λG : 0.8, σG : 0.2, πQ : 3, πE : 1, πT : 1, λSS : 2.4, σSS : 0.6, Temperature: 40
1.96	60076-5007-1001	Connection	Multi-Pin		J9	50*0.5mm-Flip-Latch	1	0.007985	0.007985	Quality Level: Level I, Number of Pins: 50		Level I	λG : 2, σG : 0.5, πQ : 3, πE : 1, πT : 1, λSS : 6, σSS : 1.5, Temperature: 40
1.97	NRS3015T2R2MNGH	Inductor	Coil	Chip/Ceramic Filter	L702,L703,L704	2.2uH-1.48A	3	0.000462	0.001386	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λG : 0.1, σG : 0.07, πQ : 3, πE : 1, πT : 1, λG : 0.1, σG : 0.07, λSS : 0.3, σSS : 0.21, Temperature: 40
1.98	HXNR4030B-4R7M	Inductor	Coil	Chip/Ceramic Filter	L3,L705	4.7uH-2A	2	0.000499	9.980442E-004	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λG : 0.1, σG : 0.07, πQ : 3, πE : 1, πT : 1, λG : 0.1, σG : 0.07, λSS : 0.3, σSS : 0.21, Temperature: 40
1.99	S16403	Inductor	Transformer	Power (> 1W)	T1	[S16403]100M POE-1 channel	1	0.023018	0.023018	Type: Power (> 1W), Quality Level: Level I		Level I	λG : 5, σG : 2, πQ : 3, πE : 1, πT : 1, λG : 5, σG : 2, λSS : 15, σSS : 6, Temperature: 40



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SI No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.100	GEP13	Inductor	Transformer	Power (> 1W)	T2	GEP13-127 μ H	1	0.023018	0.023018	Type: Power (> 1W), Quality Level: Level I		Level I	λ G: 5, σ G: 2, π Q: 3, π E: 1, π T: 1, λ G: 5, σ G: 2, λ SS: 15, σ SS: 6, Temperature: 40
1.101	0.47mH-1.8A	Inductor	Coil	Chip/Ceramic Filter	L2	0.47mH-1.8A	1	0.000581	5.806969E-004	Type: Chip/Ceramic Filter, Quality Level: Level I		Level I	λ G: 0.1, σ G: 0.07, π Q: 3, π E: 1, π T: 1, λ G: 0.1, σ G: 0.07, λ SS: 0.3, σ SS: 0.21, Temperature: 40
1.102	RS1M-F1-0002	Semiconductor	Diode	General Purpose, Si	D25	1000V-1A-SMA	1	0.000801	8.012066E-004	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.2, Rated Current: 1, Percentage: 20, Operating Voltage: 50, Rated Voltage: 1000, Percentage: 5	Rated Current: 1, Percentage: 20, Operating Voltage: 50, Rated Voltage: 1000, Percentage: 5	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.41895155, λ G: 0.33, σ G: 0.23, λ SS: 0.41476203, σ SS: 0.28907657, Temperature: 40
1.103	P00805B-LV1	Semiconductor	Diode	General Purpose, Si	D16,D17,D31,D 85,D1401,D140 3	6.5V-4KV-SMB	6	0.004517	0.027101	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.001, Rated Current: 100, Percentage: 0.001, Operating Voltage: 3.3, Rated Voltage: 6.5, Percentage: 50.7692307692	Rated Current: 100, Percentage: 0.001, Operating Voltage: 3.3, Rated Voltage: 6.5, Percentage: 50.7692307692	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.23566226, λ G: 4.6, σ G: 3.3, λ SS: 3.2521391, σ SS: 2.3330563, Temperature: 40
1.104	1SS355	Semiconductor	Diode	General Purpose, Si	D18,D21,D22,D 801	80V-0.1A-SOD323	4	0.000454	0.001818	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.01, Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Rated Current: 0.1, Percentage: 10, Operating Voltage: 5, Rated Voltage: 80, Percentage: 6.25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.31348618, λ G: 0.33, σ G: 0.23, λ SS: 0.31035132, σ SS: 0.21630546, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.105	SCD310H	Semiconductor	Diode	General Purpose, Si	D3,D24	100V-3A-2010/SMAF	2	0.001014	0.002028	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 3, Percentage: 33.333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Rated Current: 3, Percentage: 33.333333333, Operating Voltage: 12, Rated Voltage: 100, Percentage: 12	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.61672421, λ G: 0.33, σ G: 0.23, λ SS: 0.61055697, σ SS: 0.42553971, Temperature: 40
1.106	SB6100CTL-13	Semiconductor	Diode	Voltage Regulator, Zener	D15	100V-6A-DPAK	1	0.017551	0.017551	Type: Voltage Regulator, Zener, Quality Level: Level I, Actual Zener Current or Power: 1, Rated Zener Current or Power: 6, Percentage: 16.6666666667	Rated Zener Current or Power: 6, Percentage: 16.6666666667	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.44932896, λ G: 9.3, σ G: 2.8, λ SS: 12.536278, σ SS: 3.7743633, Temperature: 40
1.107	BV-SMBJ58C2H	Semiconductor	Diode	TVS Surge Suppressor	D23,D27	58V-10us/700us-85MAX-SMB	2	0.002390	0.004781	Type: TVS Surge Suppressor, Quality Level: Level I, Average Forward Current: 0.2, Rated Forward Current: 21.37, Percentage: 0.93589143659	Rated Forward Current: 21.37, Percentage: 0.93589143659	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.24102391, λ G: 2.55, σ G: 0.81, λ SS: 1.8438329, σ SS: 0.5856881, Temperature: 40
1.108	BV-SMBJ20CAI	Semiconductor	Diode	General Purpose, Si	D7	20V-10/1000us-800W-30V@8/20us-SMB	1	0.006982	0.006982	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 1, Rated Current: 30, Percentage: 3.333333333, Operating Voltage: 12, Rated Voltage: 30, Percentage: 40	Rated Current: 30, Percentage: 3.333333333, Operating Voltage: 12, Rated Voltage: 30, Percentage: 40	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.25837757, λ G: 4.6, σ G: 3.3, λ SS: 3.5656104, σ SS: 2.5579379, Temperature: 40
1.109	TUSD03FB	Semiconductor	Diode	TVS Surge Suppressor	D12,D13	3.3V-8/20us-350W-Max19V-SOD323	2	0.002327	0.004655	Type: TVS Surge Suppressor, Quality Level: Level I, Average Forward Current: 0.001, Rated Forward Current: 8, Percentage: 0.0125	Rated Forward Current: 8, Percentage: 0.0125	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.23465534, λ G: 2.55, σ G: 0.81, λ SS: 1.7951133, σ SS: 0.57021246, Temperature: 40



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SI No	Part Number	Category *	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.110	MB6F	Semiconductor	Diode	General Purpose, Si	Z1,Z2	Glass Passivated Single-Phase Bridge Rectifier	2	0.003924	0.007848	Type: General Purpose, Si, Quality Level: Level I, Operating Current: 0.4, Rated Current: 0.5, Percentage: 80, Operating Voltage: 48, Rated Voltage: 600, Percentage: 8	Rated Current: 0.5, Percentage: 80, Operating Voltage: 48, Rated Voltage: 600, Percentage: 8	Level I	π Q: 3, π E: 1, π T: 1, π S: 2.3869109, λ G: 0.33, σ G: 0.23, λ SS: 2.3630417, σ SS: 1.6469685, Temperature: 40
1.111	S8050-H-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q701	40V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.3333333333	Rated Power: 0.3, Percentage: 3.3333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.32627979, λ G: 3.44, σ G: 0.75, λ SS: 3.3672075, σ SS: 0.73412954, Temperature: 40
1.112	S8550	Semiconductor	Transistor	Field Effect, Silicon	Q702	40V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.3333333333	Rated Power: 0.3, Percentage: 3.3333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.32627979, λ G: 3.44, σ G: 0.75, λ SS: 3.3672075, σ SS: 0.73412954, Temperature: 40
1.113	S8050-H-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q2,Q5,Q9	40V-0.5A-SOT23	3	0.003921	0.011762	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.3333333333	Rated Power: 0.3, Percentage: 3.3333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.32627979, λ G: 3.44, σ G: 0.75, λ SS: 3.3672075, σ SS: 0.73412954, Temperature: 40
1.114	MMBT42L T1G	Semiconductor	Transistor	Field Effect, Silicon	Q16	300V/160V-0.5A-SOT23	1	0.004336	0.004336	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.3, Percentage: 3.3333333333	Rated Power: 0.3, Percentage: 3.3333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.32627979, λ G: 3.44, σ G: 0.75, λ SS: 3.3672075, σ SS: 0.73412954, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.115	FDS86242	Semiconductor	Transistor	Field Effect, Silicon	Q12	150V-4.1A-SOP8	1	0.010453	0.010453	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 1, Rated Power: 2.5, Percentage: 40	Rated Power: 2.5, Percentage: 40	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.78662786, λ G: 3.44, σ G: 0.75, λ SS: 8.1179995, σ SS: 1.7699127, Temperature: 40
1.116	YIS2305A-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q7	20V-5.4A-SOT23-6L	1	0.007293	0.007293	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.3, Rated Power: 1.2, Percentage: 25	Rated Power: 1.2, Percentage: 25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.54881164, λ G: 3.44, σ G: 0.75, λ SS: 5.6637361, σ SS: 1.2348262, Temperature: 40
1.117	ZM930P06T	Semiconductor	Transistor	Field Effect, Silicon	Q6	60V-3A-SOT23	1	0.008907	0.008907	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.5, Rated Power: 1.5, Percentage: 33.3333333333	Rated Power: 1.5, Percentage: 33.3333333333	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.67032005, λ G: 3.44, σ G: 0.75, λ SS: 6.9177029, σ SS: 1.5082201, Temperature: 40
1.118	B5S138W-F2-0000HF	Semiconductor	Transistor	Field Effect, Silicon	Q2601	50V-0.34A-SOT323	1	0.004697	0.004697	Type: Field Effect, Silicon, Application Type: Switch, Quality Level: Level I, Dissipated Power: 0.01, Rated Power: 0.15, Percentage: 6.6666666667	Rated Power: 0.15, Percentage: 6.6666666667	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.35345468, λ G: 3.44, σ G: 0.75, λ SS: 3.6476523, σ SS: 0.79527303, Temperature: 40
1.119	DB0049V1.0	Optical Device	Other	Single LED / LCD Segment	IR1,IR2,IR3,IR4,I R5,IR6,IR7,IR8,I R9,IR10,IR11,IR1 2,IR13,IR14,IR15 ,IR16,IR17,IR18	Night Mode	18	0.000917	0.016507	Type: Single LED / LCD Segment, Quality Level: Level I		Level I	π Q: 3, π E: 1, π T: 1, λ G: 0.25, σ G: 0.18, λ SS: 0.75, σ SS: 0.54, Temperature: 40
1.120	[HFD42/4.5-3SR(825)]-DC5V-HF	Relay	General Purpose		K1	HFD42/4.5-3SR(825)	1	0.009483	0.009483	Quality Level: Level I, Contact Current: 1, Rated Current: 4, Percentage: 25	Contact Current: 1, Rated Current: 4, Percentage: 25	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.72252735, λ G: 1.9, σ G: 1.9, λ SS: 4.1184059, σ SS: 4.1184059, Temperature: 40



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Reviewed By: Shubham Kumar (Scientist D)

Approved By: V. Jambulingam (Head Reliability)

Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.121	TSTP650K-180BH-8.4	Switch	Pushbutton		SW2	DC12V-50mA-4PIN-6*6*5mm	1	0.007842	0.007842	Quality Level: Level I, Contact Current: 0.01, Rated Current: 0.05, Percentage: 20	Contact Current: 0.01, Rated Current: 0.05, Percentage: 20	Level I	π Q: 3, π E: 1, π T: 1, π S: 0.67705687, λ G: 1.99, σ G: 1.4, λ SS: 4.0420295, σ SS: 2.8436389, Temperature: 40
1.122	K-F-24.000M-9-2020-4085-80	Miscellaneous	Crystal Oscillator		X803	24MHz $\pm$ 20ppm-(-40°C-85°C)-3.2*2.5	1	0.015628	0.015628	Oscillator Type: Quartz Controlled, Quality Level: Level I		Level I	π Q: 3, π E: 1, π T: 1, λ G: 3.2, σ G: 1.5, λ SS: 9.6, σ SS: 4.5, Temperature: 40
1.123	SF32WK327 68031T002	Miscellaneous	Crystal Oscillator		X802	32.768K $\pm$ 20ppm-(-40°C-85°C)-3.2*1.5	1	0.015628	0.015628	Oscillator Type: Quartz Controlled, Quality Level: Level I		Level I	π Q: 3, π E: 1, π T: 1, λ G: 3.2, σ G: 1.5, λ SS: 9.6, σ SS: 4.5, Temperature: 40
1.124	YT30028FB.I R-JZ-DK-	Rotating Device				Lens-Electric Zoom	2	1.000000	2.000000				
1.125	IR0109	Rotating Device				Day and Night Mode	1	1.000000	1.000000				
1.126	LMV612MM	Integrated Circuit	Analog		U701	[TP10-2-VR]-VSSOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.127	ETA5052S2G	Integrated Circuit	Analog		U1	[ETA5052S2G/E-TA5052V0S2G]-SOT23-6	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.128	LP3981AB5F j-SOT23-5	Integrated Circuit	Analog		U7,U8	[LP3981AB5F]-SOT23-5	2	0.026871	0.053741	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.129	SGM431XN3 LG	Integrated Circuit	Analog		U13,U15	[LT1431APTSLT1 G]-SOT23	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.130	TL2845BDR-8	Integrated Circuit	Analog		U12	[TL2845BDR-8/SA2845SATR/SAP2845SATR]-SOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.131	DIO54335TS T6]-SOT23-6-dioo	Integrated Circuit	Analog		U709	[DIO54335TS16]-SOT23-6	1	0.028473	0.028473	Quality Level: Level I, Number of Transistors: 456		Level I	π Q: 3, π E: 1, π T: 1, λ G: 4.8804343, σ G: 3.450467, λ SS: 14.641303, σ SS: 10.351401, Temperature: 40
1.132	ETA3485S2F]-SOT23-5-ETA	Integrated Circuit	Analog		U702,U703,U706	[ETA3485S2F]-SOT23-5	3	0.022631	0.067893	Quality Level: Level I, Number of Transistors: 456		Level I	π Q: 3, π E: 1, π T: 1, λ G: 4.8804343, σ G: 3.450467, λ SS: 14.641303, σ SS: 10.351401, Temperature: 40
1.133	ETA5053V18ODF1E]-1.8V-DFN1*1-ETA	Integrated Circuit	Analog		U710	[ETA5053V18ODF1E]-1.8V-DFN1*1	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.134	LP3981AB5F]-SOT23-5	Integrated Circuit	Analog		U707	[LP3981AB5F]-SOT23-5	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.135	TMI5302]-TSOT23-6	Integrated Circuit	Analog		U6	[TMI5302]-TSOT23-6	1	0.035814	0.035814	Quality Level: Level I, Number of Transistors: 768		Level I	π Q: 3, π E: 1, π T: 1, λ G: 6.1386501, σ G: 4.3400256, λ SS: 18.41595, σ SS: 13.020077, Temperature: 40
1.136	SD4924STR]-S08	Integrated Circuit	Analog		U10	[SD4924STR]-SOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40



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Sl No	Part Number	Category	Subcategory	Type	Reference Designator	Description	Quantity (n)	Failure Rate	n x Failure Rate	Part Data	Stress Data	Quality	π Factors
1.137	EL357N(B)(T A)-VG	Integrated Circuit	Analog		U11	[EL357N(B)(T A)(BMS)-G]-S06	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.138	TP8485E-SR	Integrated Circuit	Analog		U20	[TP8485E-SR/SIT3088EEESA]-SOP8	1	0.031289	0.031289	Quality Level: Level I, Number of Transistors: 565		Level I	π Q: 3, π E: 1, π T: 1, λ G: 5.3630921, σ G: 3.7917061, λ SS: 16.089276, σ SS: 11.375118, Temperature: 40
1.139	IMX335LQN-C	Integrated Circuit	Digital		U5	[IMX335LQN-C]-5.0MP-1/2.8-BGA88	1	0.031007	0.031007	Quality Level: Level I, Technology: CMOS, Number of Gates: 5000000		Level I	π Q: 3, π E: 1, π T: 1, λ G: 6.9374105, σ G: 2.5470825, λ SS: 20.812232, σ SS: 7.6412476, Temperature: 40
1.140	W25N01GV ZEIG	Integrated Circuit	ROM, PROM, EPROM		U1101	[W25N01GVZEI]-G]-W50N8 (8x6mm)-3V3	1	0.249449	0.249449	Quality Level: Level I, Technology: CMOS, Number of Bits: 1, Bit Units: Gb		Level I	π Q: 3, π E: 1, π T: 1, λ G: 42.756504, σ G: 30.228848, λ SS: 128.26951, σ SS: 90.686545, Temperature: 40
1.141	NT5CB256M16ER-FL	Integrated Circuit	RAM		U1001, U1002	[NT5CB256M16ER-FL]-FBGA96	2	0.099792	0.199583	Quality Level: Level I, Technology: Dynamic NIMOS/CMOS, Number of Bits: 4, Bit Units: Gb		Level I	π Q: 3, π E: 1, π T: 1, λ G: 20.379802, σ G: 13.601017, λ SS: 61.139406, σ SS: 40.80305, Temperature: 40
1.142	C5-36GW	Integrated Circuit	Microprocessor		U801	[C5-36GW]-BGA307	1	0.154317	0.154317	Quality Level: Level I, Technology: CMOS, Bus Width in Bits: 32		Level I	π Q: 3, π E: 1, π T: 1, λ G: 24.604507, σ G: 20.224777, λ SS: 73.81352, σ SS: 60.674332, Temperature: 40
1.143	[ML414H-IV01E]-3V Fuse-4A	Miscellaneous	Battery		BAT801	Rechargeable, 1mAh	1	0.104891	0.104891	Battery Type: Lithium, Quality Level: Level I		Level I	π Q: 3, π E: 1, λ G: 25, σ G: 7.5, λ SS: 75, σ SS: 22.5
1.144	Fuse-4A	Miscellaneous	Fuse		FU1	[6125SB4A250V-G]-4A-250V 6125	1	0.001964	0.001964	Fuse Type: $\leq$ 30A, Quality Level: Level I		Level I	π Q: 3, π E: 1, λ G: 0.4, σ G: 0.19, λ SS: 1.2, σ SS: 0.57



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